SANDBOX · PROFILE STUDIO — TUTORIAL 2

Using a CSV

Define an arbitrary, curved profile from a width/height trace. A CSV can describe a whole etched profile, not just a simple opening.

The CSV format

Two columns, one row per point:

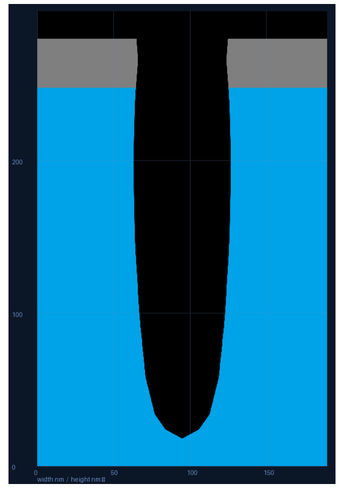
width = the full opening width (CD) at that height — a size, always ≥ 0.

height = the vertical position (nm) — may be negative (it is a position).

The opening is drawn **symmetric** about the centerline (±width/2).

width	height
0	0
22	6
36	16
48	40
56	80
62	130
64	180
62	220
58	248
60	262

example_profile.csv (a realistic etched trench)



The trace loaded as the profile

Steps

1. Prepare a CSV with columns **width,height** (see example). Save it anywhere.
2. In **Base feature**, click **“Load CSV…”** and choose your file.
3. **Placement.** By default the trace is **top-aligned** (its top sits at the stack top, cutting downward). To pin it elsewhere, set **“CSV height=0 at (nm)”** — the row where height = 0 lands that many nm from the stack bottom; negative heights sit below that line.
4. While a CSV opening is active, **Space** and **Opening depth** are ignored; **Top vacuum** still applies.
5. If the curved walls look jagged, raise **Smoothing** (Off / 2× / 4× / 8×). Then **Render** and **Save .bmp**.

Tips

• If the material stack is empty, a silicon base is added automatically so the opening is visible.

• Negative width is rejected; ragged rows / blanks are rejected with a clear message.

• Sort isn’t required — points are ordered by height internally.